

SHREYASH DHOOT

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EDUCATION

MIT World Peace University (MITWPU) Pune, India

CGPA: **8.32**

B.Tech, Electronics & Communication Engineering – AI & ML Specialisation

Expected June 2027

Coursework: OOP, DSA, Deep Learning, Image Processing, Probability & Statistics, Integral Calculus, Data Science, Data Mining, ML

Suryadatta National School, Pune

HSC 2023 – **84%**

Podar International School, Pune

CBSE Grade 10, 2021 – **93%**

EXPERIENCE

Pragya Labs, BITS Goa

Aug 2025 – Present

Research Intern – Natural Language Processing & Multi-Modal Robustness (Under Dr. Amitava Das)

- **GuardPaint** (under review, EMNLP 2026): proposed speculative safety decoding for T2I diffusion models – a plug-and-play decoding-time defense evaluated against five adversarial attack families and five architectures, achieving substantial ASR reduction with zero modification to base model weights.
- Designed a two-stage inpainter alignment pipeline (SFT → BCO); derived a **Quadratic Hinge identity guardrail** blocking pixel-smudging reward hacking; assembled an 82K image-prompt dataset with VLM-based automated labelling; extended BCO with multi-step DDIM unrolling to close the proxy/inference gap.
- **Stochastic Chaos** [arXiv:2601.07239](under review, EMNLP 2026): empirically demonstrated that deterministic LLM inference collapses rich trajectory into single brittle reasoning traces, constituting a category error in decoding policy.
- Constructed reasoning graphs over CoT benchmarks (GSM8K, SVAMP, StrategyQA) and operationalised the **Collapsed Failure Rate** (R_{coll}) metric to formally separate model incapacity from policy-induced suppression of valid reasoning.

Foresights by OmniFeed – AI-Powered Construction Site Management Platform

2025 – Present

Co-founder

- **B2B SaaS platform** targeting the construction industry; core product reduces communication gaps between site personnel and provides an end-to-end snag detection, assignment, and closure pipeline across multi-building projects.
- Built a **WhatsApp-native defect reporting pipeline** supporting voice, image, and multilingual text input; Gemini API auto-categorises reported snags into Safety, Design, Resource, and Quality with AI-generated resolution suggestions.
- Developed a **conversational assistant** enabling natural-language querying of live site data across the dashboard, and an on-demand **AI report generation** module that synthesises site intelligence into structured analytical reports.

Heera Software

Summer 2025

Machine Learning Intern – Location Intelligence & Address Verification

- Framed address-coordinate cross-verification as a **Positive-Unlabelled (PU) learning** problem over 13,510 retailer records; engineered four confidence signals (proximity, cosine text similarity, landmark similarity, address granularity) and benchmarked five approaches – nnPU, Spy+RF, anomaly ensembles, noisy-label learning, K-Means.
- Achieved **ROC-AUC 0.97, F1 0.949, and precision 0.93** using an iterative ROC-SVM framework; designed a three-class schema (Likely Valid / Inconclusive / Likely Invalid) for prioritised manual review at scale.
- Developed the **Verification Triangle** framework characterising three orthogonal failure modes (string similarity, proximity, landmark context) for interpretable fraud diagnostics.

PROJECTS

CampusCritique – Context-Aware Agentic Feedback Platform

2024

Agno Agentic Framework, AWS (Lambda, API Gateway), DynamoDB, REST APIs

- Designed and prototyped a context-aware feedback system under the university's **AI in Campus** initiative; leveraged Agentic AI and LLMs to dynamically branch follow-up questions based on prior responses, improving depth and reducing drop-off across decentralised student and faculty survey workflows.
- Architected a serverless AWS backend (Lambda + API Gateway + DynamoDB) validated through internal university stakeholder testing.

ACHIEVEMENTS

DataQuest Hackathon – Top 20 of 315 teams

- Built ML pipelines for sepsis risk prediction and credit card default detection on high-dimensional, class-imbalanced datasets; applied SMOTE, undersampling, feature selection, and XGBoost ensembles under time-constrained conditions.

SKILLS

Programming Languages: Python, C++

ML / AI Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, XGBoost, Diffusers, Agno

Research Areas: Large Language Models, Adversarial Robustness, Multimodal Safety, Diffusion Models, PU Learning